

Implement a Decision Tree classifier using the ID3 algorithm to segment customers based on their purchasing behavior using the Online Retail dataset. Analyze the tree structure and discuss the feature importance.

Tasks:

- Load and preprocess the Online Retail dataset.
- Implement Decision Tree using the ID3 algorithm.
- Visualize the decision tree and analyze feature importance.
- Discuss how the tree structure helps in understanding customer behavior.

Aim

To implement a **Decision Tree Classifier using the ID3 algorithm** to segment customers based on their purchasing behavior using the **Online Retail dataset**, visualize the decision tree, and analyze feature importance.

Algorithm

Step 1: Load the Online Retail dataset.

Step 2: Remove unnecessary or invalid records.

Step 3: Create features:

- Quantity → Number of items purchased
- UnitPrice → Price per item
- Country → Customer location
- InvoiceDate → Purchase date
- CustomerID

Step 4: Define target:

High Value Customer (1)
Low Value Customer (0)

Based on:

$$\text{Total Spend} = \text{Quantity} \times \text{UnitPrice}$$

Step 5: Train ID3 model

.

Step 6: Visualize tree

Step 7: Analyze feature importance

Code

```
# -----  
  
# 1. Import Libraries  
  
# -----  
  
import pandas as pd  
  
import numpy as np  
  
from sklearn.tree import DecisionTreeClassifier, plot_tree  
  
from sklearn.preprocessing import LabelEncoder  
  
import matplotlib.pyplot as plt  
  
  
# -----  
  
# 2. Load Dataset  
  
# -----  
  
# Download from:  
  
# https://archive.ics.uci.edu/ml/datasets/Online+Retail  
  
  
df = pd.read_excel("Online Retail.xlsx")  
  
  
# -----  
  
# 3. Data Preprocessing  
  
# -----  
  
# Remove missing values  
  
df = df.dropna(subset=['CustomerID'])  
  
  
# Remove negative/invalid values  
  
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)]
```

```
# Create Total Spend
```

```
df['TotalSpend'] = df['Quantity'] * df['UnitPrice']
```

```
# -----
```

```
# 4. Customer-Level Aggregation
```

```
# -----
```

```
customer_df = df.groupby('CustomerID').agg({
```

```
    'Quantity': 'sum',
```

```
    'UnitPrice': 'mean',
```

```
    'TotalSpend': 'sum',
```

```
    'Country': 'first'
```

```
}).reset_index()
```

```
# -----
```

```
# 5. Create Target Variable
```

```
# -----
```

```
# High Value if above median spend
```

```
threshold = customer_df['TotalSpend'].median()
```

```
customer_df['HighValue'] = (customer_df['TotalSpend'] > threshold).astype(int)
```

```
# -----
```

```
# 6. Encode Categorical Data
```

```
# -----
```

```
le = LabelEncoder()
```

```
customer_df['Country'] = le.fit_transform(customer_df['Country'])
```

```
# Features and target
```

```

X = customer_df[['Quantity', 'UnitPrice', 'Country']]
y = customer_df['HighValue']

# -----

# 7. Train Decision Tree (ID3)

# -----

model = DecisionTreeClassifier(criterion='entropy', max_depth=4)
model.fit(X, y)

# -----

# 8. Visualization

# -----

plt.figure(figsize=(20,8))

plot_tree(model,
          feature_names=X.columns,
          class_names=['Low','High'],
          filled=True)

plt.title("Customer Segmentation using Decision Tree (ID3)")
plt.show()

# -----

# 9. Feature Importance

# -----

importance = pd.Series(model.feature_importances_, index=X.columns)
print("\nFeature Importance:\n", importance.sort_values(ascending=False))

```